Alex Spigler

alex.spigler@gmail.com | alexspigler.dev | linkedin.com/in/alexspigler | github.com/alexspigler

Education

George Washington University — Expected May 2027
Dual Degree: BS in Statistics, BA in Computer Science; Minor in Mathematics — *Washington, DC*
GPA: 3.77 | Statistics GPA: 3.88

Experience

Data Analyst Intern — Summer 2026
AnswersNow — *Remote - Washington, DC*

- Authored a data-rules specification defining about 25 client and utilization metrics, resolving a cross-team cancellation-rate discrepancy so management could track performance using consistent measures.
- Proposed and completed a manager-approved competing-risks analysis of 2,121 clients using cause-specific Cox models, with time-varying and influence diagnostics. A 10-percentage-point higher Stage-1 attendance rate was associated with lower dropout hazard at day 90 (HR 0.840) and day 365 (HR 0.936), providing a basis to test early attendance support as a retention strategy.
- Built a 39-chart Mode report from 23,406 client-week observations in PostgreSQL; decomposed a 313-hour monthly decline and found 78.4% remained outside client-count and recorded-cancellation drivers, showing those factors were insufficient explanations.

Barista — Feb. 2023 – Nov. 2024
Tatte Bakery & Cafe — *Washington, DC*

- Trained 7 new employees on store operations and customer service while maintaining dual degree studies.

Projects

U.S. Unemployment Rate Forecasting | *R*

- Built a reproducible forecasting study on 942 observed monthly BLS unemployment rates (1948–2026), preserving a source-documented missing month and reserving the final 24 calendar months as holdout.
- Compared 24 diagnostically specified SARIMA models and ETS across 10 rolling windows; on a held-out 24-month period, the selected model beat naive and drift on MAE but not RMSE, exposing regime sensitivity.

Chaotic Attractors: Computational Exploration & Visualization | *Python*

- Built a pip-installable Python library (CLI + API) for searching, validating, and visualizing 9 four-parameter 2D systems (5 established forms, 4 custom variants).
- Designed a seeded, staged search that removes burn-in and validates survivors with finite-time largest Lyapunov estimates; ranked outputs by geometry and rendered density with subsampled Gaussian KDE.

Technical Skills

Languages:	Python, R, SQL, Java
Statistical Methods:	Experimental Design & Power Analysis, Bayesian Inference & MCMC, Regression Diagnostics, Survey Sampling & Weighting, Time Series & Intervention Analysis, Survival Analysis & Competing Risks
Mathematical Methods:	Linear Algebra, Dynamical Systems & Stability Analysis, Numerical ODE Methods
Libraries & Tools:	NumPy, pandas, SciPy, Matplotlib, Git, pytest, Mode Analytics, R Markdown
Databases:	PostgreSQL